

Q2. Problem K Key Persons

Time Limit: 3 seconds

There is a secret club, named No-Communication-Permitted-Club (NCPC), formed by n college students and every student has a unique ID which is a positive integer. Two students can talk if and only if their ID have a common factor bigger than 1. For convenience, we use the ID to represent the students. Two students p and q are “**relevant**” if either they can talk or they can communicate indirectly via other students. Otherwise, they are “**non-relevant**”.

Hence, two students p and q are relevant if either p and q have a common factor bigger than 1 or there is a sequence of integers $(a_1 = p, a_2, \dots, a_k = q)$ in S such that a_i and a_{i+1} for $1 \leq i \leq k-1$ have a common factor bigger than 1, where S is the set of all ID of the students. The club needs to elect some “key persons” to manage the club. A student x in S is a key person if after excluding x from S , it can cause two relevant students $p \in S$ and $q \in S$, become non-relevant in $S' = S - x$.

For example, consider $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Then, 2 and 9 are relevant because sequence $(2, 6, 9)$ exists, where 2 and 6 have a common factor 2 (>1) and 6 and 9 have a common factor 3 (>1). However, after excluding 6 from S , 2 and 9 become non-relevant in the resulting set $S' = S - 6 = \{1, 2, 3, 4, 5, 7, 8, 9\}$. Hence, 6 is a key person.

Given a set club members S , your task is to write a computer program to compute the number of key persons in S .

Technical Specification

All ID are integers at least one and at most 30000 and the number of students in each test case is no more than 1000.

Input:

The first line of the input file contains an integer, denoting the number of test cases to follow. For each test case, the club $S = \{a_1, a_2, \dots, a_n\}$ is given with the following format: The first line contains a positive integer n . In the following n lines, each line represents one integer in S such that an integer in the i^{th} line represents a_i .

Output:

For each test case, output the number of key persons in S .

Input example:

2
8
2
4
6
8
10
12
14
16
10
1
2
3
4
5
6
7
8
9
10

Output Example:

0
2